We will prove that the answer is affirmative.

We will denote p/q-Dehn surgery on a knot K in S^3 by $K(p/q)$. It is known that if $K(p/q)$ is reducible then $q = 1$ [GL], and K is either hyperbolic or a cable knot (this includes torus knots) [S]. Furthermore, if K is a cable knot then $K(p)$ has at most two prime summands [GL]. It is also known that if $K(p)$ has more than two prime summands then it has exactly three, and is of the form $L_1 \# L_2 \# Q$, where L_1 and L_2 are lens spaces and Q is a homology sphere (see [H, Corollary 5.3]).

The following is the main result of [HLZ].

Theorem 1. (Hom, Lidman, and Zufelt). *If K is a hyperbolic L-space knot and $K(p)$ is reducible then $p = 2g(K) - 1$.*

If M is a connected 3-manifold, $\tilde{M}$ will denote the universal abelian covering of M. If G is a finitely generated group, $\rho(G)$ will denote the minimal number of generators of G.

The following is contained in [GS, Proof of Proposition 1.4].

Lemma 2. (González-Acuña and Short). *If K is a fibered knot of genus g then $\rho(\pi_1(\widetilde{K(p/q)})) \leq 2g$.*

Proof. Let $\pi(K) = \pi_1(S^3 \setminus K)$. Since $\pi(K)$ maps onto $\pi_1(K(p/q))$, the commutator subgroup $\pi(K)'$ maps onto $\pi_1(K(p/q))' = \pi_1(\widetilde{K(p/q)})$. Since $\pi(K)'$ is free of rank $2g$, we have $\rho(\pi_1(\widetilde{K(p/q)}))) \leq \rho(\pi(K)') = 2g$. □

Lemma 3. *Let M and N be connected 3-manifolds with $H_1(M) \cong \mathbb{Z}_m$ and $H_1(N) \cong \mathbb{Z}_n$. Then*

$$\widetilde{M\#N} \cong (\#_{(m-1)(n-1)} S^1 \times S^2) \# (\#_n \tilde{M}) \# (\#_m \tilde{N})$$

Proof. Let M_0 (N_0) denote M (N) with an open 3-ball removed. Then $M\#N \cong M_0 \cup S^2 \times I \cup N_0$. Under the covering projection $\widetilde{M\#N} \to M\#N$, the inverse image of M_0 is n disjoint copies of $\tilde{M}_0 = \tilde{M}$ with m disjoint open 3-balls removed, the inverse image of N_0 is m disjoint copies of $\tilde{N}_0 = \tilde{N}$ with n disjoint open 3-balls removed, and the inverse image of $S^2 \times I$ is mn copies of $S^2 \times I$, joining each copy of $\tilde{M}_0$ to each copy of $\tilde{N}_0$. If Γ is the graph with vertices in 1-1 correspondence with the copies of $\tilde{M}_0$ and $\tilde{N}_0$ and edges in 1-1 correspondence with the copies of $S^2 \times I$, then $\chi(\Gamma) = (m + n) - mn$, and hence the first Betti number $b_1(\Gamma) = 1 - \chi(\Gamma) = (m - 1)(n - 1)$. This gives the stated form of $\widetilde{M\#N}$. □

Corollary 4. *Let M and N be lens spaces $L(m, s)$ and $L(n, t)$, respectively, and let Q be a homology 3-sphere. Then*

$$\widetilde{M\#N\#Q} \cong (\#_{(m-1)(n-1)} S^1 \times S^2) \# (\#_{mn} Q)$$

Proof. Since $\tilde{M} \cong S^3$ and $\widetilde{N\#Q} \cong \#_n Q$, the result follows from Lemma 3. □

Proof that the answer to the Question is affirmative.

Suppose for contradiction that K is an L-space knot of genus g such that $K(p)$ has more than two prime summands. Then K is hyperbolic, so $p = 2g - 1$ by Theorem 1. Also, $K(p) \cong L(m,s)\#L(n,t)\#Q$, where $H_1(Q) = 0$. Therefore $p = mn$ and

$$\widetilde{K(p)} \cong (\#_{(m-1)(n-1)} S^1 \times S^2)\#(\#_p Q)$$

by Corollary 4. Hence

$$\pi_1(\widetilde{K(p)}) \cong F_{(m-1)(n-1)} * (*_p \pi_1(Q))$$

By Grushko's Theorem

$$\rho(F_{(m-1)(n-1)} * (*_p \pi_1(Q))) = (m-1)(n-1) + p\rho(\pi_1(Q))$$

Since $H_1(Q) = 0$ and Q is not S^3, $\rho(\pi_1(Q)) \geq 2$. Therefore

$$\rho(\pi_1(\widetilde{K(p)})) \geq (m-1)(n-1) + 2p = 3mn + 1 - (m+n)$$

By [N] K is fibered, and hence by Lemma 2, $3mn+1-(m+n) \leq 2g = mn+1$, giving $2mn-(m+n) \leq 0$. But $2mn-(m+n) = (mn-1)+(m-1)(n-1) > 0$ since $m, n \geq 2$, a contradiction.

References

[GS] F. González-Acuña and H. Short, *Knot surgery and primeness*, Math. Proc. Camb. Phil. Soc. **99** (1986), 89–102.

[GL] C. McA. Gordon and J. Luecke, *Only integral Dehn surgeries can yield reducible manifolds*, Math. Proc. Camb. Phil. Soc. **102** (1987), 97–101.

[HLZ] J. Hom, T. Lidman and N. Zufelt, *Reducible surgeries and Heegaard Floer homology*, Math. Res. Lett. **22** (2015), 763–788.

[H] J. Howie, *A proof of the Scott-Wiegold conjecture on free products of cyclic groups*, J. Pure Appl. Algebra **173** (2002), 167–176.

[N] Y. Ni, *Knot Floer homology detects fibered knots* , Invent. Math. **170** (2007), 577–608.

[S] M. Scharlemann, *Producing reducible 3-manifolds by surgery on a knot*, Topology **29** (1990), 481–500.